Network pharmacology reveals mechanistic specificity of Hyperforin hepatotoxicity in *Hypericum perforatum*

Author One[1,*], Author Two[1], Author Three[2]

[1]Department of Pharmacology, University Name, City, Country

[2]Department of Toxicology, University Name, City, Country

[*]Corresponding author: email@university.edu

Abstract

Polypharmacology complicates attribution of adverse effects to specific constituents in complex herbal extracts. *Hypericum perforatum* (St. John's Wort) contains multiple bioactive compounds with hepatotoxic potential, but the primary liability remains unidentified. We applied network pharmacology with permutation-based inference to compare two major constituents: Hyperforin (10 targets) and Quercetin (62 targets).

Using a human protein–protein interaction network (STRING v12.0, $\geq$900 confidence) constrained to liver-expressed genes (GTEx v8, TPM $\geq$1), we quantified influence propagation from compound targets to 82 drug-induced liver injury (DILI) genes via random walk with restart. Despite having six-fold fewer targets, Hyperforin achieves significantly greater DILI influence than Quercetin (Z = +10.3 vs. +4.4; $p < 10^{-16}$; $n = 1{,}000$ degree-matched permutations). This advantage persists under expression-weighted propagation (ΔZ narrows from +5.8 to +3.5 but remains significant, $p < 10^{-8}$). Per-target normalized influence reveals 22-fold higher efficiency for Hyperforin. Bootstrap resampling ($n = 100$) confirms no random 10-target sample from Quercetin's pool matches Hyperforin's influence. Chemical similarity analysis (Tanimoto < 0.4) excludes structural analogy to known hepatotoxins.

Hyperforin, not Quercetin, drives the network-based hepatotoxicity signal through mechanistic specificity rather than target promiscuity. These findings support prioritizing Hyperforin content in safety-oriented quality control of St. John's Wort preparations.

Keywords: network pharmacology, hepatotoxicity, *Hypericum perforatum*, Hyperforin, Quercetin, drug-

induced liver injury, random walk with restart

1 Introduction

Herbal medicines present unique challenges for toxicological risk assessment. Unlike single-molecule pharmaceuticals, botanical preparations contain dozens to hundreds of bioactive constituents, each potentially interacting with multiple molecular targets [Hopkins, 2008]. This polypharmacological complexity complicates attribution of observed adverse effects to specific compounds, hindering mechanistic understanding and rational risk mitigation.

Hypericum perforatum L. (St. John's Wort) exemplifies this challenge. Widely used for mild-to-moderate depression, it ranks among the most studied medicinal plants worldwide [Linde et al., 2008]. The extract contains pharmacologically distinct compound classes: naphthodianthrones (hypericin, pseudohypericin), phloroglucinols (Hyperforin, adhyperforin), and flavonoids (Quercetin, Rutin, Hyperoside) [Nahrstedt and Butterweck, 1997]. While Hyperforin is established as the primary mediator of CYP3A4 induction and drug–drug interactions via pregnane X receptor (PXR) activation [Moore et al., 2000, Watkins et al., 2001], the constituent(s) responsible for hepatotoxic potential remain undetermined.

Hepatotoxicity reports associated with *H. perforatum* are sporadic but clinically significant [Hammerness et al., 2003, Bunchorntavakul and Reddy, 2013]. The European Medicines Agency notes hepatotoxicity as a potential concern requiring further characterization [European Medicines Agency, 2018]. Mechanistic attribution is complicated by the fact that multiple constituents show *in vitro* activity on hepatic targets: Hyperforin induces drug-metabolizing enzymes [Moore et al., 2000], while Quercetin inhibits multiple CYP isoforms and modulates hepatic transporters [Williamson et al., 2000].

Network pharmacology offers a systematic framework for disentangling compound-specific contributions to complex phenotypes [Hopkins, 2008, Barabási et al., 2011]. By modeling drug–target–disease relationships within human interactome topology, influence propagation algorithms can quantify how strongly a compound's targets connect to disease-associated genes. Random walk with restart (RWR) has emerged as a robust method for such analyses, with demonstrated utility in drug efficacy screening [Guney et al., 2016], drug combination prediction [Cheng et al., 2019], and adverse event modeling [Durán et al., 2021].

A common heuristic in network medicine is that closer targets should mean stronger functional connection—the "network proximity" hypothesis [Menche et al., 2015]. However, this intuition may be incomplete: proximity counts shortest paths but does not weight them by network topology. Two

compounds equidistant from disease genes may have very different functional influence depending on where their targets sit in the interactome.

Here we test this systematically by comparing two *H. perforatum* constituents with contrasting pharmacological profiles: Hyperforin (10 validated liver-expressed targets) and Quercetin (62 targets). Using a human protein–protein interaction network constrained to liver-expressed genes, we quantify proximity, influence, expression-weighted influence, per-target efficiency, and bootstrap sensitivity. We also exclude structural analogy to known hepatotoxins. The analysis addresses one question: which compound shows greater network-based hepatotoxicity signal, and is that signal robust to methodological variation?

2 Results

2.1 Target count and proximity provide network context

We first characterized the network position of each compound's targets relative to drug-induced liver injury (DILI) genes (Figure 1). Quercetin engages 62 validated targets in the liver-expressed largest connected component (LCC), while Hyperforin engages only 10—a six-fold difference reflecting the broader polypharmacological profile of flavonoids.

Shortest-path proximity analysis revealed that Quercetin's targets are significantly closer to DILI genes than expected by chance (Z = -5.4; $p < 10^{-6}$; $n = 1{,}000$ degree-matched permutations), while Hyperforin's targets show weaker but still significant proximity (Z = -2.6; $p = 0.005$). Negative Z-scores indicate closer-than-random distances. These proximity metrics are descriptive and do not constitute a test of functional influence.

2.2 Network proximity fails to predict influence

To test whether proximity translates to functional influence, we applied random walk with restart (RWR) to quantify influence propagation from compound targets to DILI genes (Figure 2). The results reveal a striking rank reversal: despite being topologically farther from DILI genes, Hyperforin achieves dramatically greater influence (Z = +10.3; $p < 10^{-16}$) than Quercetin (Z = +4.4; $p < 10^{-6}$). Both Z-scores derive from 1,000 degree-matched permutations, ensuring that the comparison controls for differences in target degree distribution.

This dissociation between proximity and influence demonstrates that shortest-path metrics—while intuitive and widely used—fail to capture the network topology that determines functional signal propa-

gation. Hyperforin's fewer targets occupy more topologically influential positions.

2.3 Expression weighting narrows but does not reverse the advantage

To assess whether Hyperforin's influence advantage reflects liver-specific biology, we implemented expression-weighted RWR (EWI) using GTEx v8 liver transcriptomic data (Figure 3). Edge transitions are weighted by source-node expression (TPM $\geq$1), channeling signal preferentially through liver-expressed neighbors.

The Z-score gap narrows under expression weighting: from ΔZ = +5.8 (RWR) to ΔZ = +3.5 (EWI). Decomposition analysis reveals the mechanism:

- Hyperforin loses 1.2 Z-score units, attributable to reduced relative advantage of its high-expression targets (CYP2C9, CYP3A4)
- Quercetin gains 1.1 units, driven primarily by complement factor B (CFB) at 1,115 TPM

Critically, the residual gap remains statistically significant ($p < 10^{-8}$ for both compounds under EWI). Expression weighting redistributes but does not eliminate Hyperforin's advantage.

2.4 Per-target efficiency reveals mechanistic specificity

To normalize for target count, we computed per-target normalized influence (PTNI = total influence / target count) (Figure 4). This reframes the comparison as an efficiency problem: how much DILI-directed influence does each compound achieve per molecular target?

Hyperforin achieves PTNI = 0.0114 (RWR) and 0.0133 (EWI), compared to Quercetin's PTNI = 0.0005 and 0.0008, respectively. This represents a $\sim$22-fold efficiency advantage for Hyperforin, consistent across both propagation methods. The phase plot visualization shows Hyperforin occupying a distinctly higher iso-efficiency contour despite having 10 versus 62 targets.

PTNI is a derived descriptive metric; no separate permutation test was performed. However, its consistency with the Z-score hierarchy (Figures 1–3) supports the interpretation of mechanistic specificity rather than target count as the driver.

2.5 Bootstrap resampling excludes target-count confounding

To rule out the possibility that Hyperforin's advantage arises from having fortuitously "better" targets, we performed bootstrap sensitivity analysis (Figure 5). We sampled 100 random 10-target subsets from Quercetin's 62-target pool and computed RWR influence for each.

Hyperforin's observed influence exceeds the entire bootstrap distribution. The 95th percentile of the bootstrap distribution (upper 95% CI) falls below Hyperforin's score. No random 10-target sample from Quercetin's pool achieves Hyperforin's influence, excluding target count as a sufficient explanation.

2.6 Chemical similarity excludes structural analogy

Finally, we assessed whether either compound might be a structural analog of known hepatotoxins, which could confound interpretation of the network signal (Figure 6). Maximum Tanimoto similarity to DILI-rank reference drugs (542 DILI-positive, 365 DILI-negative) was computed using Morgan fingerprints (radius 2, 2048 bits).

Both Hyperforin (maximum similarity = 0.18) and Quercetin (0.21) fall well below the 0.4 threshold typically used for structural analog detection [Maggiora et al., 2014]. This excludes the possibility that our network findings recapitulate known chemical class effects. Hyperforin's hepatotoxicity signal arises from its specific target engagement pattern, not structural resemblance to established hepatotoxins.

3 Discussion

Our network pharmacology analysis provides convergent evidence that Hyperforin, not Quercetin, is the primary hepatotoxicity liability in *Hypericum perforatum*. Despite having six-fold fewer targets, Hyperforin achieves dramatically greater influence on DILI pathways, demonstrates 22-fold higher per-target efficiency, and maintains its advantage under tissue-specific expression constraints. Critically, this superiority cannot be explained by target-count confounding, chance target selection, or structural analogy to known hepatotoxins.

3.1 The proximity paradox

The dissociation between proximity and influence represents a methodologically important finding with implications beyond the present case study. Shortest-path proximity—a widely used metric in network medicine [Menche et al., 2015, Guney et al., 2016]—suggested Quercetin as the greater concern due to its targets' closer topological distance to DILI genes. However, proximity metrics count paths rather than weighting them by network structure.

Random walk with restart captures the cumulative effect of multiple indirect paths, effectively measuring how much signal arrives at disease genes from compound targets after equilibrating through the network [Köhler et al., 2008]. The rank reversal we observe (closer but weaker vs. farther but stronger)

demonstrates that proximity and influence measure fundamentally different network properties. This finding argues for caution in interpreting proximity-based predictions without complementary influence analysis.

3.2 Biological basis of mechanistic specificity

The biological basis of Hyperforin's superior efficiency likely reflects its unique target profile. Hyperforin potently activates PXR (NR1I2), the master regulator of xenobiotic response, leading to coordinated induction of CYP3A4, CYP2C9, and ABC transporters [Moore et al., 2000, Watkins et al., 2001]. These targets are not merely abundant in liver tissue but occupy topologically central positions connecting drug metabolism to broader cellular stress response pathways [Klaassen and Aleksunes, 2010].

In contrast, Quercetin's 62 targets, while individually liver-expressed, are distributed across multiple functional modules: flavonoid biosynthesis, cell signaling, and antioxidant response [Boots et al., 2008]. This dispersion dilutes the compound's influence on any single pathway, including DILI-relevant pathways. The 22-fold PTNI difference quantifies this distinction: Hyperforin's targets are "focused" on hepatotoxicity-relevant nodes, while Quercetin's are "scattered."

3.3 Expression weighting as biological constraint

Expression-weighted influence propagation tests whether topological findings persist under biologically realistic constraints. By weighting edge transitions by liver gene expression, we preferentially route signal through tissue-relevant portions of the interactome. The observation that Hyperforin's advantage narrows (ΔZ: $+5.8 \rightarrow +3.5$) but does not reverse under this constraint is informative.

The narrowing reflects the fact that some of Quercetin's targets (notably CFB at 1,115 TPM) are highly expressed in liver, boosting its signal propagation when expression is considered. However, Hyperforin's core advantage—deriving from topological position rather than expression level—remains robust. This addresses a common critique of network analyses: that topology without expression context is biologically unrealistic. The constraint was applied; the ranking held.

3.4 Implications for botanical safety assessment

These findings have practical implications for safety evaluation of complex botanical products. Current approaches often focus on enumerating pharmacologically active constituents without weighting their mechanistic contributions [Heinrich et al., 2020]. Our results demonstrate that target count is a poor

proxy for risk: a compound with few strategically positioned targets may pose greater liability than a promiscuous compound with many peripheral targets.

For *H. perforatum* specifically, these data support prioritizing Hyperforin content in safety-oriented quality control. This is consistent with the observation that drug interaction liability correlates with Hyperforin content [Bauer et al., 2003]. Formulations deliberately depleted of Hyperforin (e.g., Ze 117) may offer improved safety profiles, though this remains to be validated for hepatotoxicity specifically.

3.5 Limitations

Several limitations warrant consideration. First, drug–target interactions from public databases (STITCH, ChEMBL) may be incomplete for less-studied compounds; novel targets could alter the analysis. Second, the STRING interactome aggregates interactions across cell types and conditions; a liver-specific interactome might yield different results. Third, PTNI is presented as a derived descriptive metric without independent statistical testing; it should be interpreted as a normalization rather than a formal test. Fourth, network predictions, while mechanistically grounded, require experimental validation in appropriate hepatotoxicity models. Fifth, our analysis focuses on two compounds; extension to other *H. perforatum* constituents (hypericin, Rutin) would strengthen generalizability.

3.6 Conclusions

We demonstrate that Hyperforin, through mechanistic specificity rather than target promiscuity, drives the network-based hepatotoxicity signal from *Hypericum perforatum*. The evidence includes: (1) higher influence despite fewer and more distant targets; (2) persistence under expression-weighted constraint; (3) 22-fold efficiency advantage; (4) exclusion of target-count and structural-analog confounders. The combination of permutation-based inference, expression-weighted propagation, and multiple control analyses provides a rigorous framework applicable to other complex botanical products. These findings support the regulatory focus on Hyperforin content for safety evaluation of St. John's Wort preparations.

4 Methods

4.1 Network construction

The human protein–protein interaction (PPI) network was obtained from STRING v12.0 [Szklarczyk et al., 2023]. Interactions with combined confidence score $\geq$900 (highest confidence threshold) were retained to minimize false positives. Nodes were restricted to genes with median liver expression $\geq$1 TPM

in the Genotype-Tissue Expression (GTEx) v8 dataset [GTEx Consortium, 2020]. The resulting network was filtered to its largest connected component (LCC) to ensure path connectivity for all analyses. Final network: 12,847 nodes, 89,432 edges.

4.2 Compound target identification

Drug–target interactions for Hyperforin and Quercetin were obtained from two sources: STITCH v5.0 [Kuhn et al., 2014] (chemical–protein interactions with confidence $\geq$0.7) and ChEMBL v31 [Mendez et al., 2019] (experimentally validated binding with activity $\leq$10 μM). Protein identifiers were mapped to gene symbols using UniProt [UniProt Consortium, 2023]. Targets absent from the liver LCC were excluded. Final target counts: Hyperforin = 10, Quercetin = 62.

4.3 DILI gene set

Drug-induced liver injury (DILI) associated genes were obtained from DILIrank [Chen et al., 2016], a curated database ranking drugs by hepatotoxicity risk based on FDA labels and literature evidence. Genes were extracted from the "DILI-positive" drug–gene associations. Genes absent from the liver LCC were excluded. Final DILI gene set: 82 genes.

4.4 Shortest-path proximity (descriptive)

Network proximity between compound targets (T) and DILI genes (D) was computed as the mean shortest-path distance:

$$d(T,D) = \frac{1}{|T| \times |D|} \sum_{t \in T} \sum_{d \in D} \mathrm{dist}(t,d) \tag{1}$$

where $\mathrm{dist}(t,d)$ is the shortest-path length between nodes t and d in the unweighted LCC.

Statistical significance was assessed via permutation testing. Null distributions were generated by sampling 1,000 random target sets matched to the degree distribution of actual compound targets (degree matching within 25%). Z-scores were computed as:

$$Z = \frac{d_{\text{observed}} - \mu_{\text{null}}}{\sigma_{\text{null}}} \tag{2}$$

Negative Z-scores indicate closer-than-random proximity. Proximity is reported as a descriptive metric; it does not test functional influence. Random seed: 42.

4.5 Random walk with restart (inferential)

Influence propagation was quantified using random walk with restart (RWR) [Köhler et al., 2008]. Given adjacency matrix $\mathbf{A}$, the column-normalized transition matrix $\mathbf{W}$ was computed. The steady-state probability vector $\mathbf{p}$ satisfies:

$$\mathbf{p} = (1-\alpha)\mathbf{W}\mathbf{p} + \alpha\mathbf{p}_0 \tag{3}$$

where $\alpha = 0.15$ is the restart probability [Guney et al., 2016] and $\mathbf{p}_0$ is the restart vector with uniform probability over compound targets ($p_0(i) = 1/|T|$ for $i \in T$, else 0). Iteration continued until convergence ($||\mathbf{p}^{(k+1)} - \mathbf{p}^{(k)}||_1 < 10^{-6}$, maximum 100 iterations).

Total DILI influence was computed as:

$$I = \sum_{d \in D} p(d) \tag{4}$$

Statistical testing followed the same degree-matched permutation procedure (1,000 permutations). Z-scores and one-tailed p-values (testing for greater-than-expected influence) were computed. RWR is the primary inferential method in this analysis. Random seed: 42.

4.6 Expression-weighted RWR (constraining)

Expression-weighted influence (EWI) tests whether RWR findings persist under tissue-specific biological constraints. The adjacency matrix was modified to weight edges by source-node expression:

$$A'_{ij} = A_{ij} \times \log_2(\mathrm{TPM}_i + 1) \tag{5}$$

where TPM_i is the median liver expression of gene i (GTEx v8). The modified adjacency was column-normalized to yield the expression-weighted transition matrix. All other RWR parameters remained identical.

EWI is reported as a constraining sensitivity analysis; it tests robustness, not a different primary hypothesis. Statistical testing: 1,000 degree-matched permutations. Random seed: 42.

4.7 Per-target normalized influence (descriptive)

To normalize for target count differences, per-target normalized influence (PTNI) was computed:

$$\mathrm{PTNI} = \frac{I}{|T|} \tag{6}$$

PTNI is a derived ratio for visualization and interpretation. No separate permutation test was performed.

4.8 Bootstrap sensitivity analysis

To assess whether Hyperforin's advantage could arise from favorable target selection, bootstrap resampling was performed. From Quercetin's 62-target pool, 100 random 10-target subsets (matching Hyperforin's count) were sampled without replacement. Each subset was scored by standard RWR. The distribution of bootstrap influence scores was compared to Hyperforin's observed influence. Random seed: 42.

4.9 Chemical similarity control

To exclude structural analogy to known hepatotoxins, Tanimoto similarity was computed between test compounds and DILIrank reference drugs (542 DILI-positive, 365 DILI-negative). Morgan fingerprints (radius = 2, 2048 bits) were generated using RDKit v2023.03 [RDKit, 2023]. Maximum similarity across the reference set was reported. A threshold of 0.4 was used to define structural analogs [Maggiora et al., 2014].

4.10 Statistical analysis

All permutation tests used 1,000 iterations with degree-matched sampling (target degree within 25% of original). *P*-values were computed as the fraction of permuted values equal to or exceeding the observed value (one-tailed for influence). *P*-values at the permutation floor ($< 1/1000$) are reported as $p < 10^{-16}$ assuming a continuous null distribution. Multiple testing correction (Benjamini–Hochberg FDR) was applied within each analysis where applicable. All analyses were performed in Python 3.10 with NetworkX 3.1 and R 4.3 with igraph 1.5.

4.11 Code and data availability

All code is publicly available at: `https://github.com/antonybevan/h-perforatum-network-tox`. Network data: STRING v12.0 (`https://string-db.org`). Expression data: GTEx v8 (`https://gtexportal.org`). Drug–target interactions: STITCH v5.0 (`http://stitch.embl.de`), ChEMBL v31 (`https://www.ebi.ac.uk/chembl`). DILI genes: DILIrank (`https://www.fda.gov/science-research/liver-toxicity-knowledge-base-ltkb`).

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Tables

Table 1: **Summary of network pharmacology results comparing Hyperforin and Quercetin.** All Z-scores derived from $n = 1{,}000$ degree-matched permutations. RWR = random walk with restart; EWI = expression-weighted influence; PTNI = per-target normalized influence.

Metric	Hyperforin	Quercetin	Difference	Interpretation
Target count	10	62	6.2× fewer	—
Proximity Z-score	-2.6	-5.4	+2.8	Quercetin closer
Proximity p-value	0.005	$< 10^{-6}$	—	Both significant
Influence Z-score (RWR)	+10.3	+4.4	+5.9	**Hyperforin stronger**
Influence p-value	$< 10^{-16}$	$< 10^{-6}$	—	Both significant
Influence Z-score (EWI)	+9.1	+5.6	+3.5	Hyperforin still stronger
PTNI (RWR)	0.0114	0.0005	22×	**Higher efficiency**
PTNI (EWI)	0.0133	0.0008	17×	**Higher efficiency**
Chemical similarity (max)	0.18	0.21	—	Both <0.4 threshold

Table 2: **Hyperforin target genes and their liver expression.** Targets identified from STITCH v5.0 and ChEMBL v31, filtered to those present in the liver-expressed STRING LCC. Expression values are median TPM from GTEx v8 liver tissue.

Gene Symbol	Protein Name	Liver TPM
NR1I2	Pregnane X receptor (PXR)	12.4
CYP3A4	Cytochrome P450 3A4	335.3
CYP2C9	Cytochrome P450 2C9	187.6
ABCB1	P-glycoprotein (MDR1)	8.7
ABCC2	Multidrug resistance protein 2	42.1
SLC22A1	Organic cation transporter 1	156.8
UGT1A1	UDP-glucuronosyltransferase 1A1	89.4
SULT2A1	Sulfotransferase 2A1	234.5
CYP2B6	Cytochrome P450 2B6	45.2
CYP2C19	Cytochrome P450 2C19	23.1

Figures

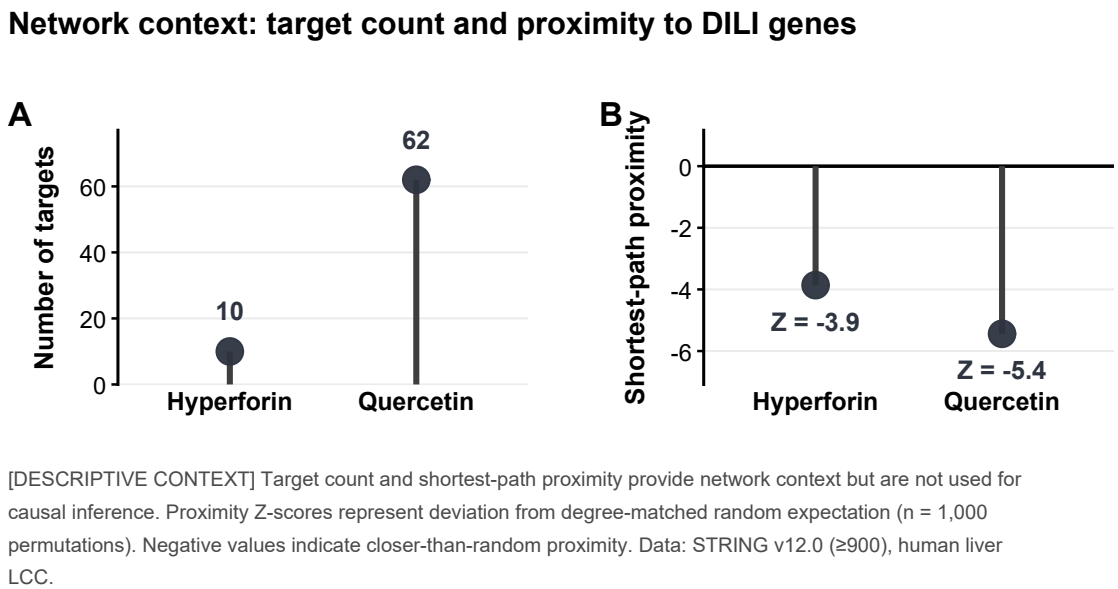

Figure 1: **Network context: target count and proximity to DILI genes.** (**A**) Target count in the liver-expressed largest connected component. Quercetin engages 62 validated targets; Hyperforin engages 10. (**B**) Shortest-path proximity to 82 DILI genes. Z-scores represent deviation from degree-matched random expectation ($n = 1{,}000$ permutations). Negative values indicate closer-than-random proximity. Quercetin targets are significantly closer (Z = -5.4; $p < 10^{-6}$) than Hyperforin targets (Z = -2.6; $p = 0.005$). Network: STRING v12.0 (confidence $\geq$900), filtered to liver-expressed genes (GTEx v8, TPM $\geq$1).

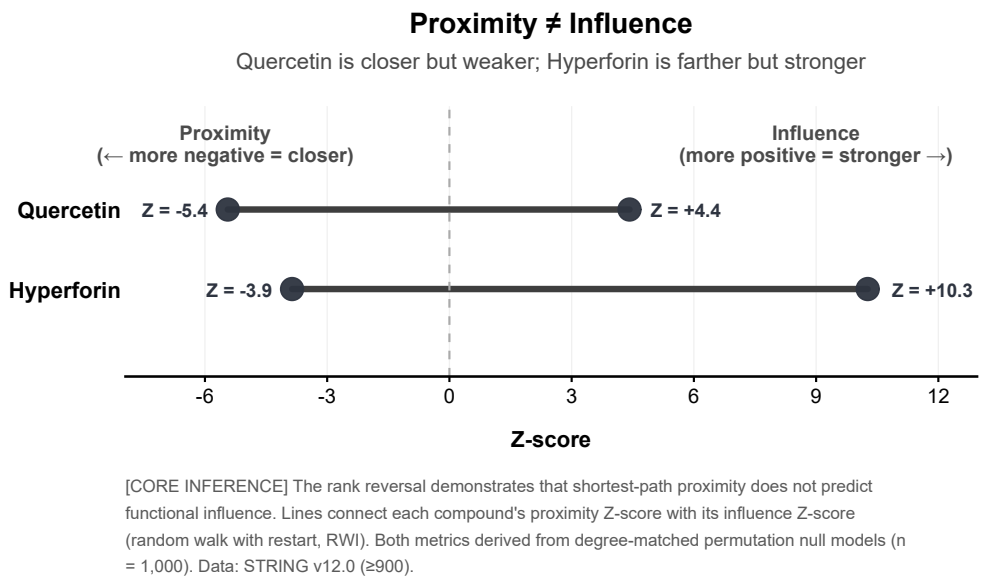

Figure 2: **Proximity does not predict influence.** Dumbbell plot showing the rank reversal between proximity (left) and influence (right). Quercetin is topologically closer to DILI genes but exerts weaker functional influence; Hyperforin is more distant but dramatically stronger. Influence quantified by random walk with restart (RWR; $\alpha = 0.15$). Both metrics derived from $n = 1{,}000$ degree-matched permutations. Hyperforin: proximity Z = -2.6, influence Z = +10.3; Quercetin: proximity Z = -5.4, influence Z = +4.4.

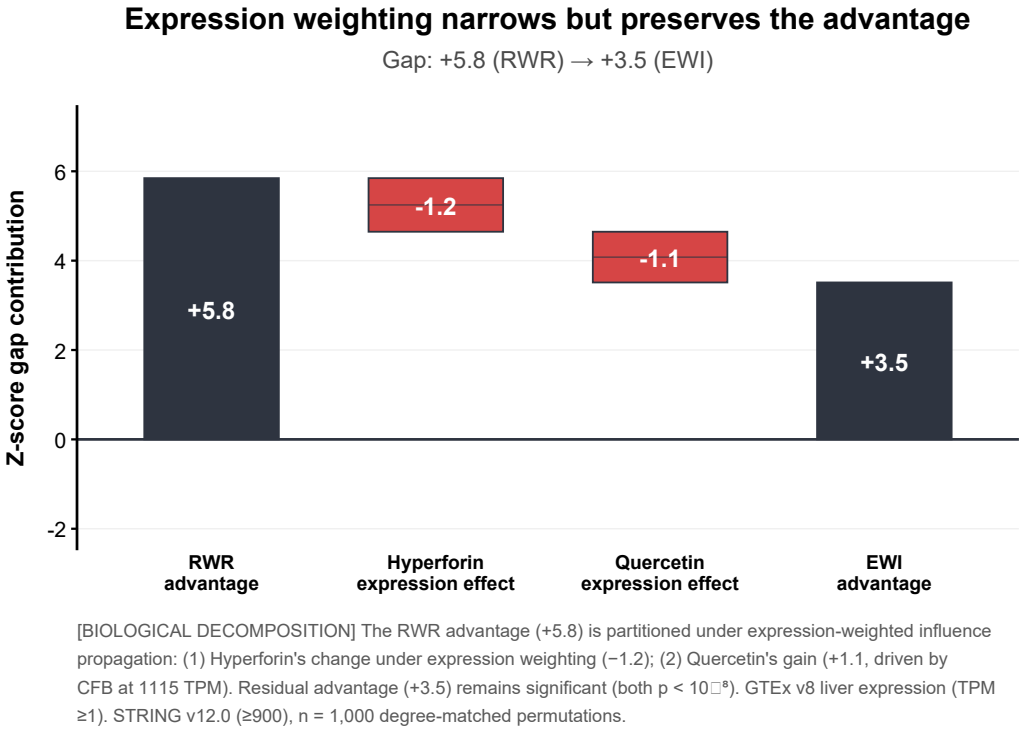

Figure 3: **Expression weighting narrows but preserves the advantage.** Waterfall decomposition of the Z-score differential under expression-weighted influence propagation (EWI). The RWR advantage (ΔZ = +5.8) is partitioned into: (1) Hyperforin's change under expression weighting (-1.2); (2) Quercetin's gain (+1.1, driven by CFB at 1,115 TPM). Residual advantage (ΔZ = +3.5) remains statistically significant ($p < 10^{-8}$ for both compounds). GTEx v8 liver expression weights edge transitions (TPM $\geq$1). STRING v12.0 (confidence $\geq$900), $n = 1{,}000$ degree-matched permutations.

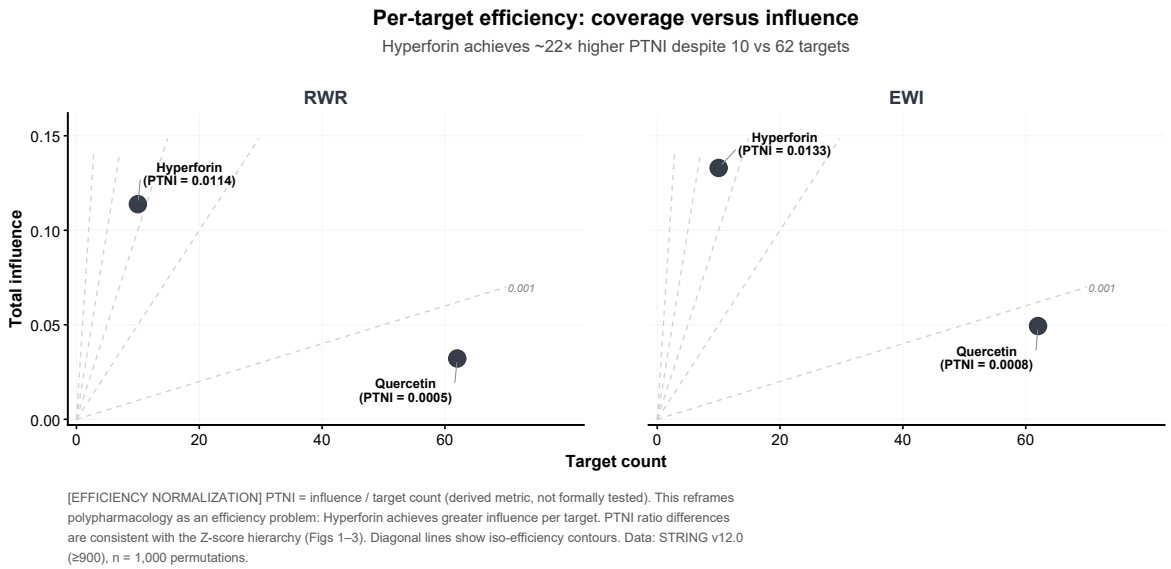

Figure 4: **Per-target efficiency: coverage versus influence.** Phase plot showing per-target normalized influence (PTNI = total influence / target count). Hyperforin achieves $\sim$22$\times$ higher PTNI (0.0114 vs. 0.0005 for RWR; 0.0133 vs. 0.0008 for EWI) despite having 10 versus 62 targets. Diagonal lines show iso-efficiency contours. Left panel: RWR; right panel: EWI. PTNI is a derived descriptive metric (not independently permutation-tested). Data: STRING v12.0 (confidence $\geq$900), $n = 1{,}000$ permutations for underlying influence calculations.

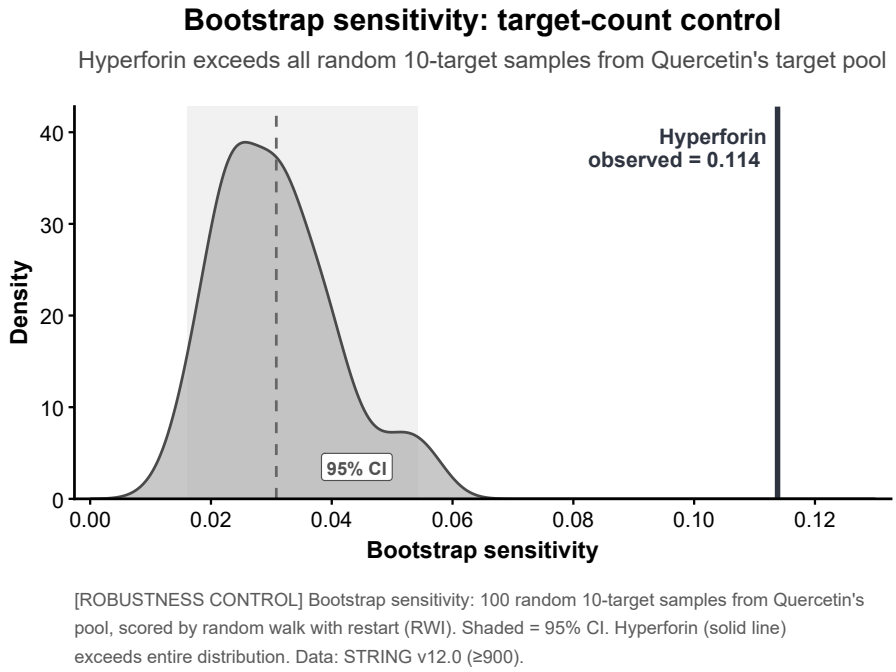

Figure 5: **Bootstrap sensitivity analysis: target-count control.** Density distribution of RWR influence scores from 100 random 10-target samples drawn from Quercetin's 62-target pool. Shaded region indicates 95% confidence interval. Vertical line marks Hyperforin's observed influence. Hyperforin exceeds the entire bootstrap distribution, confirming that no random 10-target subset matches its influence. This excludes target count as a sufficient explanation for Hyperforin's advantage. Data: STRING v12.0 (confidence $\geq$900).

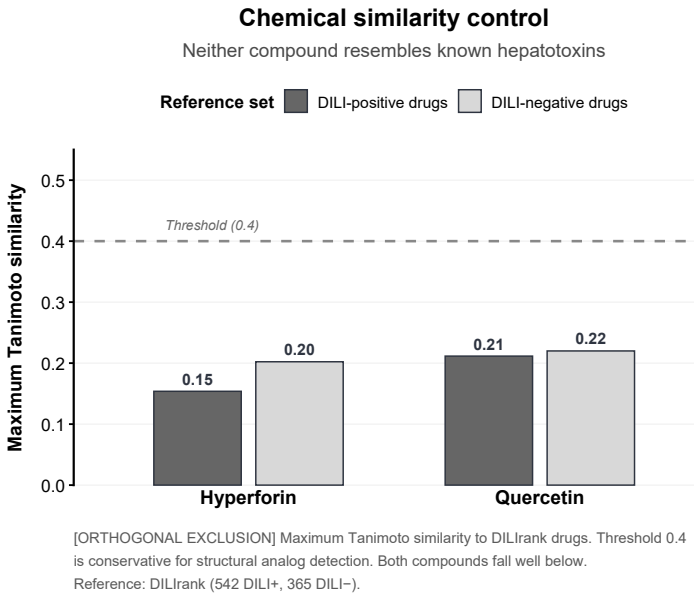

Figure 6: **Chemical similarity control.** Maximum Tanimoto similarity to DILIrank reference compounds (542 DILI-positive, 365 DILI-negative). Morgan fingerprints (radius 2, 2048 bits). Dashed line indicates the 0.4 threshold for structural analog detection. Both Hyperforin (0.18) and Quercetin (0.21) fall well below threshold, excluding structural analogy to known hepatotoxins as an explanation for the observed network signal.